
Group08-Git Push The Beats

Melody Dive
Software Development Plan (Small Project)
Version <1.0>

Melody Dive	Version: 1.0
Software Development Plan (Small Project)	Date: 30/10/2025
MD-PP-001-V1.0	

Revision History

Date	Version	Description	Author
<dd/mmm/yy>	<x.x>	<details>	<name>

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Software Development Plan (Small Project)

1. Introduction

The introduction of the Software Development Plan provides an overview of the entire document. It includes the purpose, scope, definitions, acronyms, abbreviations, references, and overview of this Software Development Plan.

Purpose

The purpose of the *Software Development Plan* is to gather all information necessary to control the project. It describes the approach to the development of the software and is the top-level plan generated and used by managers to direct the development effort.

The following people use the *Software Development Plan*:

- The **project manager** uses it to plan the project schedule and resource needs, and to track progress against the schedule.
- **Project team members** use it to understand what they need to do, when they need to do it, and what other activities they are dependent upon.

1.1 Scope

This Software Development Plan describes the overall plan to be used by the Melody Dive project, including deployment of the product.

The plans as outlined in this document are based upon the product requirements as defined in the Vision Document.

1.2 Overview

This Software Development Plan contains the following information:

Project Overview — provides a description of the project's purpose, scope, and objectives. It also defines the deliverables that the project is expected to deliver.

Project Organization — describes the organizational structure of the project team.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

This project aims to create a music streaming website that helps users search for and listen to songs. It also includes various features, such as allowing users to like and save their favorite tracks, share songs with others, etc. Our group chose this project because we are all music lovers who understand what users truly need in a music website. We aim to create an enjoyable listening experience for everyone.

Objectives:

- Delivering a web application including frontend UI, backend services (API, user auth, data storage), the implementation of all specified features.
- Delivering documentation (vision, architecture, user guide, ...).

2.2 Assumptions and Constraints

- Budget: This is a student project with no formal budget, limiting the use of paid services, software, or assets.
- Staff: A team including five members. The team has limited formal experience in developing a full-scale website, which may impact the development timeline and complexity of features.
- Equipment: Internet connection is required to ensure communication between members and

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project devices must have sufficient capacity and working memory.

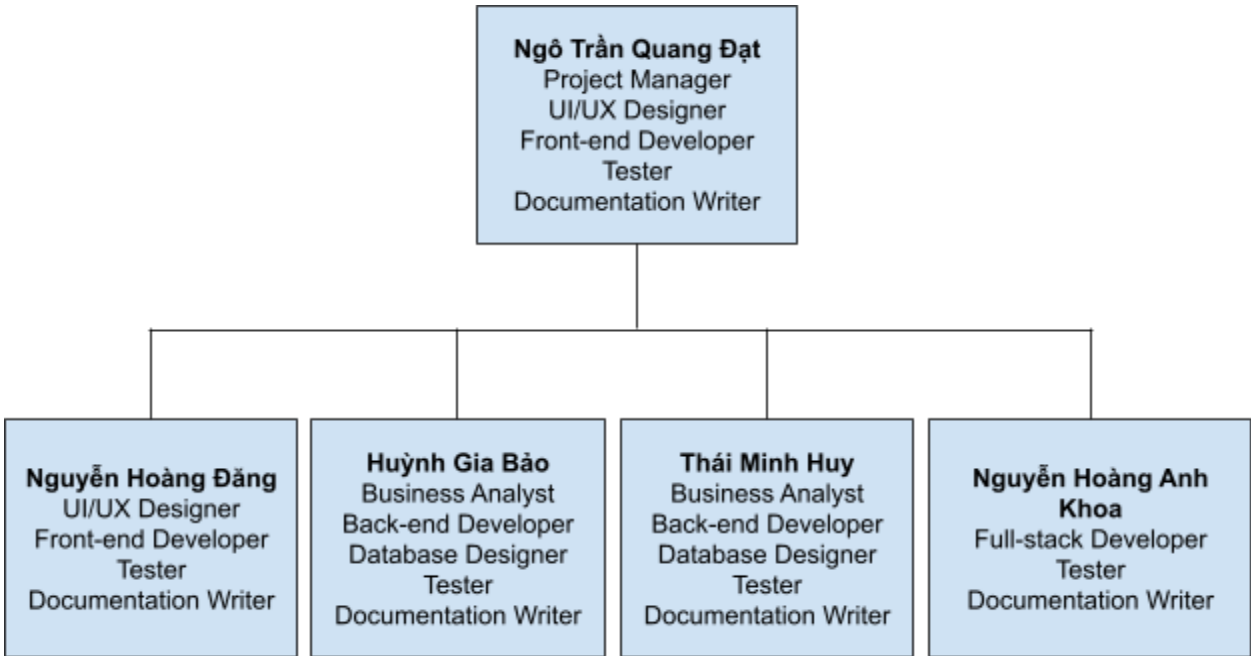
- Schedule: The project must be completed within the specified time, adhering to the sprint deadlines outlined in the project plan.

2.3 Project Deliverables

Deliverables are delivered towards the end of the iteration, as specified in section 4.2 *Project Plan*. The final deliverable is a website that meets the requirements along with documentation.

3. Project Organization

3.1 Organizational Structure



3.2 Roles and Responsibilities

Person	Role
Ngô Trần Quang Đạt	Project Manager UI/UX Designer, Front-end Developer Tester Documentation Writer
Nguyễn Hoàng Đăng	UI/UX Designer, Front-end Developer Tester Documentation Writer
Huỳnh Gia Bảo	Business Analyst Back-end Developer/Database Designer Tester Documentation Writer

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Thái Minh Huy	Business Analyst Back-end Developer/Database Designer Tester Documentation Writer
Nguyễn Hoàng Anh Khoa	Full-stack Developer Tester Documentation Writer

4. Management Process

4.1 Project Estimates

4.2 Project Plan

The whole process consists of 5 sprints, which are presented in detail below.

4.2.1. Sprint 1

Start - end date: 26/10/2025 - 8/11/2025

Tasks:

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				29/10/25												
				8/11/25												
				1/1/00												
					Oct 27, 2025				Nov 3, 2025				Nov 10,			
					27	28	29	30	31	1	2	3	4	5	6	7
					M	T	W	T	F	S	S	M	T	W	T	F
Sprint 1																
Write project plan	All members	29/10/25	2/11/25													
Learn Python basic and advanced	Bao, Huy, Khoa	29/10/25	6/11/25													
Learn HTML/CSS	Khoa, Dat, Dang	29/10/25	6/11/25													
Write vision document	All members	3/11/25	6/11/25													
Write weekly report	Quang Dat	29/10/25	7/11/25													
Review documents	All members	7/11/25	7/11/25													

Achievements after Sprint 1:

- Finished some sections of essential documents, including project plan, vision document.
- Members learned Python and HTML/CSS.

4.2.2. Sprint 2

Start - end date: 10/11/2025 - 22/11/2025

Tasks:

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4.3 Project Monitoring and Control

4.3.1 Requirements Management

The requirements for this system are captured in the Vision document. Requested changes to requirements are captured in Change Requests, and are approved as part of the Configuration Management process.

4.3.2 Reporting and Measurement

Updated cost and schedule estimates, and metrics summary reports, will be generated at the end of each iteration.

The Minimal Set of Metrics, as described in the RUP [Guidelines: Metrics](#), will be gathered on a weekly basis. These include:

Earned value for completed tasks. This is used to re-estimate the schedule and budget for the remainder of the project, and/or to identify need for scope changes.

Total defects open and closed – shown as a trend graph. This is used to help estimate the effort remaining to correct defects.

Acceptance test cases passing – shown as a trend graph. This is used to demonstrate progress to stakeholders.

In addition, overall costs will be monitored against the project budget.

4.3.3 Risk Management

Risks will be identified in Inception Phase using the steps identified in the RUP for Small Projects activity “Identify and Assess Risks”. Project risk is evaluated at least once per iteration and documented in this table. The risks of the greatest magnitude are listed first in the table.

Risk Ranking (High, Medium, Low)	Risk Description and Impact	Mitigation Strategy and/or Contingency Plan
High	Every member has fewer formal experience about developing a Website, both Front-end and Back-end.	Allocate time for team members to learn essential tools and technologies early; assign tasks according to individual strengths.
High	Copyright infringement from hosting music.	Find a way to ask the author for permission if possible or note “This is only for academics, not for profit”.
Medium	Miss deadlines	Set clear communication rules in the group, remind team members by using Jira or Trello for tasks, and announce to them in Messenger that deadlines are coming.
Medium	Personal issues such as illness or other external commitments during the project.	Team members have to announce to the group if they have personal issues so that the Project Manager will arrange time for that member to do some tasks beside the project.
Medium	Semantic search (AI feature) poor accuracy.	This feature, our team will do as best as we can to improve it better. We can’t make this feature perfect but try as best as we can.
Low	Personal conflicts happen between members.	Team members have to solve conflict problems by themselves first. If it can’t be solved, Project Manager has to be the center to solve that problem.
Low	Problems about cross-platform when	Have responsive design when working on computer browsers and mobile phone browsers.

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	browsing on mobile phone and web browser.	
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4.3.4 Configuration Management

Appropriate tools will be selected which provide a database of Change Requests and a controlled versioned repository of project artifacts.

All source code, test scripts, and data files are included in baselines. Documentation related to the source code is also included in the baseline, such as design documentation. All customer deliverable artifacts are included in the final baseline of the iteration, including executables.